Project Proposal: When Should Onion Farmers Sell Their Crop?

Oct 23, 2026 · INDUSTRY 4.0 PROJECT

Using price forecasting and optimisation to reduce farmer losses

Team **Kanda Code** proposes to build a sell-or-store decision model for onion farmers. It forecasts monthly onion prices at Lasalgaon, then uses linear programming to choose how much to sell each month after harvest, net of storage losses.

Item	Detail
Course	Industry 4.0 (MS4001)
Purpose	Project topic approval
Team name	Kanda Code
Case study	Lasalgaon APMC, Nashik, Maharashtra
Report due	End of Week 8

Team members

Name	Roll number	Email
Raghuram Devarayi	21f1005431	21f1005431@ds.study.iitm.ac.in
Parashmani Datta	21f2000987	21f2000987@ds.study.iitm.ac.in
Aaryan Choudhary	23f2003700	23f2003700@ds.study.iitm.ac.in
Kathan Desai	24f1002390	24f1002390@ds.study.iitm.ac.in
Yash Arabhavi	22f3001882	22f3001882@ds.study.iitm.ac.in

All five members contribute equally (20% each).

1. Problem statement

Onion farmers sell most of their crop at the seasonal price low because stored onions rot, and they have no data-backed way to decide when selling later pays.

In May 2026 farmers were selling onions at Rs 5–7 per kg while the all-India average retail price was Rs 25.36 per kg (Outlook Business). At Lasalgaon, farmers have received Rs 7–10 per kg against a cultivation cost of Rs 22–25 per kg (Down To Earth).

Storage is the trade-off. ICAR trials found that ventilated storage loses 54.9% of onions over 120 days, while cold storage loses only 5.0–7.1% (ICAR). A farmer group must therefore decide each month whether the expected price rise beats the expected rot. This project builds that decision.

2. Why we chose this problem

We chose it because it is a real, recurring national problem with public data, a clear decision-maker and a decision that analytics can improve.

- **It recurs every year and costs real money.** The government procured about 6.4 lakh tonnes of onion for buffer stocks in 2023–24 and set a 5 lakh tonne target for the rabi 2024 crop (Krishi Jagran).
- **Storage, not supply, is where value is lost.** In 2019 NAFED lost almost 53% of a 57,372 tonne buffer held in traditional storage (ThePrint). The same trade-off faces every farmer.
- **Onion is the hardest case.** Grains store well and have support prices; tomatoes cannot be stored at all. Onion stores for 4–6 months but rots fast, and its price swings sharply, so the sell-or-store choice is genuinely difficult.
- **Lasalgaon is the benchmark market.** It is Asia's largest onion market, its price is quoted nationally, and it has long daily price and arrival records.
- **It links analytics to Industry 4.0.** The gap between ventilated (54.9% loss) and controlled storage (5–7% loss) lets us value smart, monitored storage in rupees per tonne.
- **It is new for this course.** None of the previous batches' project titles addresses agricultural storage or selling decisions.

3. Research questions and hypotheses

We will answer three questions, each tied to a testable hypothesis.

#	Research question	Hypothesis	How we test it
RQ1	How strongly do market arrivals drive the Lasalgaon onion price?	H1: Price falls significantly as daily arrivals rise.	Regression of price on arrivals and month; sign and significance of the arrivals coefficient
RQ2	Does storing part of the rabi crop beat selling it all at harvest, after storage losses?	H2: An optimised storage plan earns more than selling everything at harvest, even with ventilated-storage losses.	LP plan vs sell-at-harvest baseline, backtested on actual 2023 and 2024 prices
RQ3	How much is low-loss (cold or IoT-monitored) storage worth to a farmer group?	H3: Low-loss storage raises net revenue per tonne by a material margin over ventilated storage.	Same LP under scenarios B and C; difference in net revenue per tonne

4. Model formulation

The decision-maker is a farmer producer organisation (FPO) holding a rabi harvest of Q tonnes in April, choosing how much to sell in each month from April (t = 1) to December (t = 9).

Objective: maximise net revenue, the forecast price times tonnes sold, minus storage cost.

Decision variables: x_t = tonnes sold in month t; s_t = tonnes held in storage at the end of month t.

Constraints: the stock balance shrinks by the monthly loss rate; storage cannot exceed capacity; a minimum quantity is sold at harvest to repay crop loans; all stock is sold by December.

Maximise $Z = \Sigma\ (\hat{p}_t \times x_t - h \times s_t)$, summed over months t = 1 to 9 (April to December)

Constraint	Formula	Meaning
Harvest month	$s_1 = Q - x_1$	Whatever is not sold in April goes into storage

Constraint	Formula	Meaning
Stock balance	$s_t = (1 - \delta) \times s_{t-1} - x_t$	This month's stock = last month's stock, minus rot, minus what we sell
Storage capacity	$s_t \leqslant C$	Stock can never exceed storage space
Minimum harvest sale	$x_1 \geqslant L$	Sell at least L tonnes in April to repay crop loans
Clear by December	$s_9 = 0$	All stock is sold by the end of the season
Non-negativity	$x_t \geqslant 0$, $s_t \geqslant 0$	No negative sales or stock

Parameter	Meaning	Source
$\hat{p}_t$	Forecast Lasalgaon price in month t (Rs/tonne)	Forecasting model, Section 6
δ	Monthly storage loss rate	ICAR storage trials
h	Storage cost (Rs/tonne/month)	Stated assumption with sensitivity
C	Storage capacity (tonnes)	Scenario setting
L	Minimum sale at harvest (tonnes)	Stated assumption
Q	Harvest held by the FPO (tonnes)	Stated assumption

Storage scenarios

Scenario	Storage	Monthly loss δ	Basis
A	None (C = 0)	–	Today's typical small farmer: sell all at harvest
B	Ventilated kanda chawl	about 18%	54.9% lost over 120 days, compounded monthly
C	Cold or IoT-monitored storage	about 1.8%	7.1% lost over 120 days (DOGR cold storage)

We treat the FPO as a price-taker, since its volume is small against Lasalgaon's daily arrivals; we will check this against the arrivals data.

5. Data

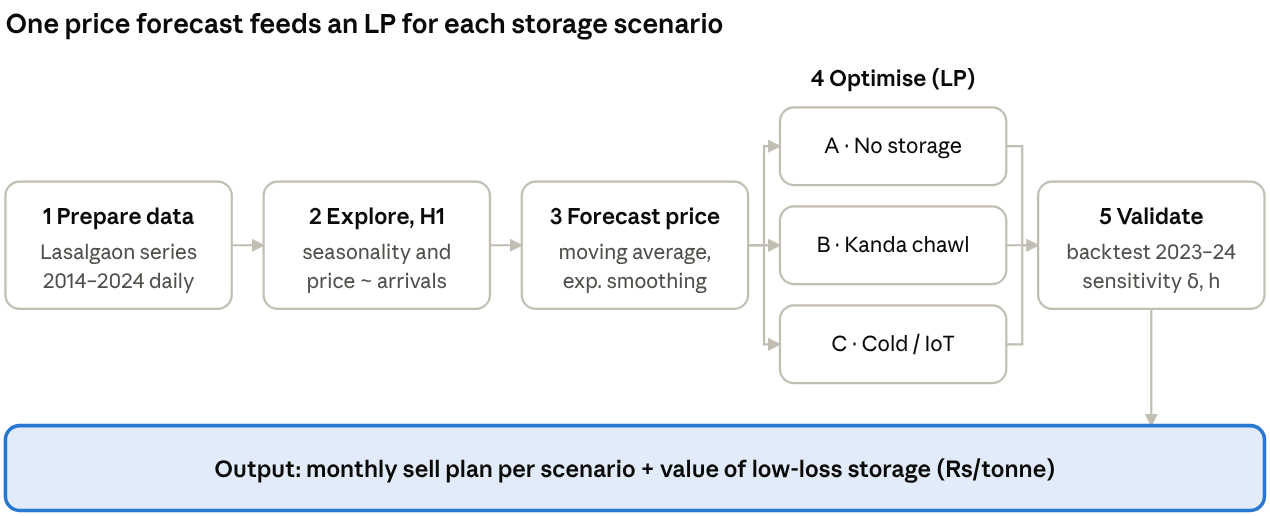
All data is public secondary data from official or peer-published sources; no personal data is used.

Dataset	What it contains	Used for
Spatio-temporal onion price dataset, 2014–2024 (Shelar, Hajare and Mangarule, Mendeley Data, 2025, CC BY 4.0)	Daily onion prices and arrivals by state, district, market and variety, compiled from Agmarknet; plus daily weather and event features	Main dataset: Lasalgaon price series, price–arrivals regression, forecasting and backtesting
Agmarknet (Directorate of Marketing and Inspection)	Official daily mandi prices and arrivals	Spot-checking the main dataset against the original source
ICAR onion storage trials	Storage losses over 120 days by storage type	Monthly loss rates δ for scenarios B and C
Published cost references (to be cited)	Storage cost per tonne, cultivation cost	Cost parameters, with sensitivity ranges

Our first task is to confirm that the Lasalgaon series in the main dataset has few gaps across 2014–2024.

6. Methodology

We forecast prices once, solve the same LP under each storage scenario, and validate every plan on prices the model has not seen.



method · 5 steps, 3 storage scenarios

1. **Prepare data.** Filter the Lasalgaon series, aggregate daily prices and arrivals to months, and document gaps and how we treat them.
2. **Explore and test H1.** Plot seasonal price patterns and harvest-time crashes; regress price on arrivals with month effects.
3. **Forecast prices.** Compare moving average, exponential smoothing and regression forecasts, trained on 2014–2022 and tested on 2023–2024; choose by forecast error (MAPE).
4. **Optimise.** Solve the LP in Excel Solver, cross-checked in Python (PuLP), for scenarios A, B and C.
5. **Validate.** Backtest each plan on actual 2023 and 2024 prices against a sell-at-harvest baseline (H2), measure the value of low-loss storage (H3), and run sensitivity on loss rate, storage cost, capacity and minimum harvest sale.

7. How we will make it research-grade

The final report will follow the standards of an applied operations research paper, within the course's report format.

- **Literature first.** A short review of onion price volatility, post-harvest storage losses and storage-timing models, so our model builds on published work.
- **Out-of-sample testing.** Forecasts are judged on 2023–2024 data the model never saw, never on the data used to fit it.

- **A real benchmark.** Every plan is compared with what farmers do today, selling at harvest, using actual historical prices.
- **Every assumption sourced or stress-tested.** Loss rates come from ICAR trials; cost figures are cited and varied in a sensitivity table, so conclusions do not hinge on one guess.
- **Generalisability.** The model needs only a price series and a loss rate; if time allows, we will rerun it on a second storable crop (potato) to show it transfers.
- **Reproducibility.** One shared notebook and Excel model, with a data dictionary and fixed train/test split, so anyone can rerun every number.
- **Honest limitations.** We will state what the model ignores, such as quality grading, price impact of large sellers and credit constraints.

8. Course alignment and expected outcomes

The project uses only standard Business Analytics and Industry 4.0 methods, applied to one clear business decision.

Concept from the courses	Where we use it
Exploratory data analysis and visualisation	Seasonality, volatility and harvest-time price crashes
Regression	Effect of arrivals on price (H1); a forecasting candidate
Time-series forecasting (moving average, exponential smoothing)	Monthly price forecasts $\hat{p}_t$
Linear programming	The month-by-month sell-or-store plan
Inventory and perishability	Stock balance with a monthly loss rate
Industry 4.0: data-driven decisions, IoT-monitored storage	Scenario C and the rupee value of smart storage

Expected outcomes

- A recommended monthly selling plan for each storage scenario.
- The revenue gain over selling at harvest, backtested on 2023 and 2024 prices.
- The value of moving from ventilated to low-loss storage, in rupees per tonne.

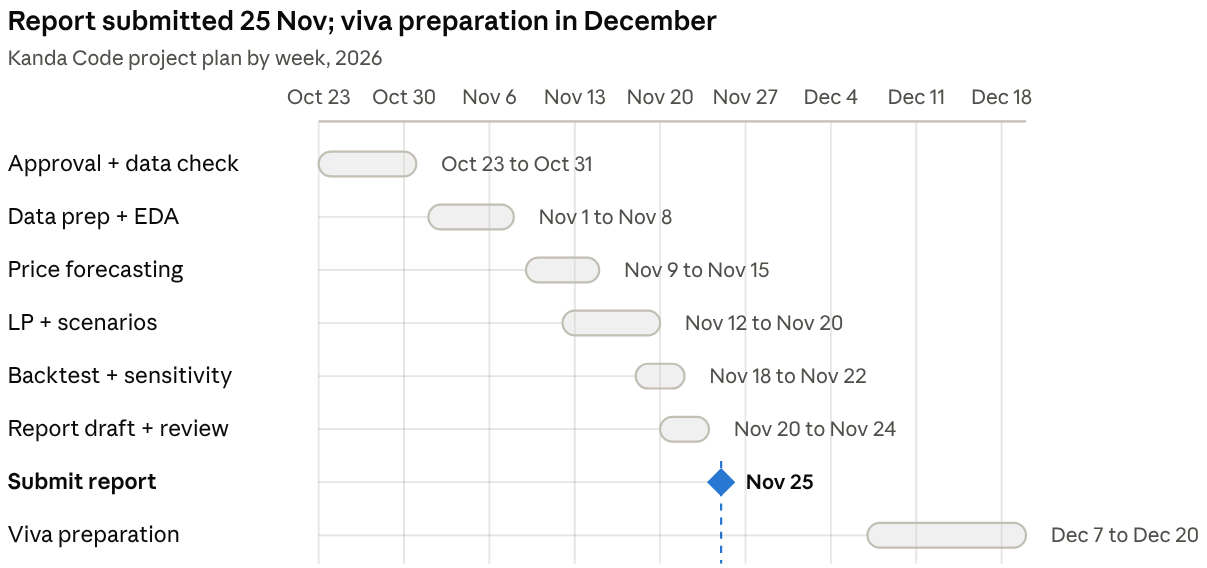
Managerial implications

- **FPOs:** when to sell and how much to hold, for their own storage.
- **Government buffer agencies:** when buying and releasing stock pays off, and why storage type matters.
- **Storage investors and policy:** the payback case for cold and IoT-monitored onion storage.

9. Team plan and timeline

Each workstream has one lead and one reviewer; all five members contribute equally and every member will be able to explain the data and the model in the viva.

Workstream	Responsibilities
Coordination	Weekly call, deadlines, submissions, communication with instructors
Data	Download, clean and document the Lasalgaon series; check against Agmarknet
Forecasting and EDA	Seasonality, H1 regression, forecast models and accuracy
Optimisation	LP in Excel Solver and PuLP, scenarios, backtest and sensitivity
Report and presentation	Literature review, report in the course format, 15-minute presentation



project plan · approval to viva preparation

The lighter week around Quiz 1 (15 Nov) is covered by the overlap between forecasting and LP work; the viva date will be set by the course team.

References

1. Shelar, R., Hajare, A. and Mangarule, S. (2025). A Comprehensive Spatio-Temporal Dataset for Agricultural Price Forecasting: Integrated Market, Weather, and Event Features (2014–2024). Mendeley Data, V1. DOI 10.17632/ds9jmxp9zy.1.
2. Directorate of Marketing and Inspection. Agmarknet: daily prices and arrivals. Government of India.
3. Indian Council of Agricultural Research. Onion Cold Storage – Potential to Stabilize Price Fluctuation.
4. Outlook Business (15 May 2026). Why farmers dump crops despite rising food prices.
5. Down To Earth (25 November 2022). Onion price crash: Maharashtra farmers to protest.
6. ThePrint (20 November 2019). NAFED wasted over 30,000 MT onions, more than half of its buffer stock.
7. Krishi Jagran (updated 26 March 2024). Govt directs NCCF and NAFED to procure 5 lakh tonnes of onion for buffer stock.